

SINGAPORE CHANGI INTL, Singapore

WMO: 486980

Lat: 1.3678N

Lon: 103.9826E

Elev: 22

StdP: 14.68

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	73.9	74.9	69.3	107.8	84.0	70.0	110.4	83.4	17.5	85.0	16.3	85.3	4.2	330	0.362

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	10.4	91.8	79.3	91.3	79.2	90.0	79.1	81.9	87.2	81.5	86.8	81.0	86.2	8.6	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.7	159.8	85.0	80.2	157.1	84.5	79.2	151.8	83.7	45.9	87.8	45.2	87.7	44.8	87.2	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.1	13.5	12.0	71.7	93.9	0.8	1.3	71.1	94.9	70.6	95.6	70.1	96.3	69.5	97.3		
(5)			70.5	83.1	1.0	0.8	69.7	83.7	69.1	84.1	68.6	84.6	67.8	85.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	83.0	81.4	82.5	83.4	83.9	84.3	83.9	83.3	83.1	83.2	83.1	82.3	81.5		
	DBStd	2.16	1.99	1.93	2.08	1.85	2.00	2.18	2.15	2.13	1.95	1.85	1.69	1.88		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD50	12040	975	909	1034	1017	1064	1017	1032	1025	995	1026	969	977		
	CDD65	6565	510	489	569	567	599	567	567	560	545	561	519	512		
	CDH74	71546	4768	4882	6069	6395	7201	6810	6613	6388	6201	6193	5213	4813		
(13)	CDH80	24824	1298	1541	2129	2376	2923	2697	2437	2248	2192	2163	1550	1270		
(14)	Wind	WSAvg	4.8	5.9	6.5	5.0	3.6	4.0	4.5	5.6	5.8	5.0	3.8	4.4		
(15) Precipitation	PrecAvg	80.9	8.7	5.2	6.3	5.7	6.3	5.4	5.6	5.7	5.4	6.4	9.8	11.8		
	PrecMax	113.7	23.7	16.5	17.6	12.1	14.6	10.5	12.0	12.9	17.3	16.7	21.1	30.2		
	PrecMin	44.1	0.6	0.0	0.9	1.0	1.7	0.9	0.7	1.8	0.9	0.4	2.9	2.5		
	PrecStd	16.9	5.9	4.1	3.8	2.7	2.7	2.4	2.6	2.4	3.3	3.6	4.4	6.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.9	91.9	93.3	93.1	93.1	91.8	90.8	91.2	91.4	91.8	91.4	90.0		
		MCWB	78.3	78.8	79.3	79.7	80.6	79.4	79.2	78.5	77.9	77.8	78.2	77.9		
	2%	DB	88.3	90.0	91.4	91.5	91.0	89.7	89.7	89.8	89.8	90.0	89.7	88.6		
		MCWB	78.4	78.3	78.7	79.9	80.4	79.7	79.2	78.6	78.5	78.3	78.7	78.5		
	5%	DB	87.7	88.9	89.9	89.9	90.0	89.7	88.9	88.2	89.0	89.4	88.2	87.6		
		MCWB	78.4	78.2	78.7	79.7	80.2	79.5	79.2	78.9	78.7	78.5	79.0	78.8		
	10%	DB	86.1	87.6	88.4	88.9	89.2	88.2	87.8	87.6	87.8	87.9	87.1	86.1		
		MCWB	77.8	78.1	78.8	79.8	80.2	79.6	79.2	78.9	78.9	78.8	78.9	78.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.6	80.7	81.7	82.3	82.8	82.2	81.7	81.4	81.4	81.4	81.0	80.9		
		MCDB	86.4	87.6	87.8	88.3	88.1	87.6	86.3	86.3	86.5	86.5	86.3	86.2		
	2%	WB	79.8	79.9	80.7	81.5	82.1	81.5	80.8	80.5	80.6	80.5	80.4	80.0		
		MCDB	85.2	86.3	86.9	87.3	87.1	86.6	85.7	85.5	85.8	85.8	85.8	85.3		
	5%	WB	79.1	79.2	80.0	80.9	81.6	80.9	80.1	80.0	80.0	80.0	79.8	79.4		
		MCDB	84.3	85.1	85.8	86.5	86.6	85.9	85.1	84.9	85.1	85.1	84.9	84.4		
	10%	WB	78.7	78.8	79.6	80.3	81.0	80.3	79.8	79.6	79.6	79.5	79.2	78.9		
		MCDB	83.6	84.4	85.0	85.7	85.9	85.3	84.8	84.4	84.5	84.4	84.0	83.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.0	11.1	11.7	11.4	10.4	9.8	9.4	9.5	10.0	10.9	10.8	10.0		
		MCDBR	11.5	12.4	12.8	12.3	11.2	10.4	9.8	10.1	10.7	11.9	12.1	11.7		
	5% WB	MCWBR	4.2	3.9	4.1	4.6	4.5	4.3	4.1	4.1	4.3	4.5	4.6	4.6		
		MCWBR	10.4	11.3	11.8	11.4	10.5	9.9	9.5	9.7	10.0	11.0	11.2	10.6		
(40) Clear-Sky Solar Irradiance	taub	taub	0.470	0.481	0.489	0.486	0.474	0.485	0.477	0.519	0.550	0.577	0.483	0.471		
		taud	2.299	2.251	2.236	2.246	2.289	2.260	2.290	2.148	2.057	1.962	2.260	2.305		
	Ebn at Noon	Ebn at Noon	273	272	268	263	259	252	256	251	248	243	267	271		
		Edh at Noon	43	46	47	45	42	42	41	49	55	61	45	43		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1444	1672	1669	1616	1494	1412	1405	1428	1426	1423	1341	1311		
	RadStd	RadStd	121	149	146	102	73	101	110	82	96	107	86	99		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (6 neighbors)	+0.67	+0.16	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Nomenclature: See separate page